

Fail-closed validation funnel

1. Deterministic construction

canonical order; untouched cases byte-identical

2. Structural certificate

indices, positive area, closed oriented 2-manifold

3. Geometric certificate

symmetric vertex Hausdorff $\leq 0.05 \times$ AABB diagonal

4. Perceptual proxy

six axial views, normal/depth SSIM, 1024^2

5. Hidden validation

one isolated branch per probe; archive Accepted lineage

A candidate is submitted only after every local gate passes; a runtime branch reverts if its transaction cannot be verified.